Tshwane University of Technology
We empower people

INSTRUCTIONS TO CANDIDATES

1. All exam rules stated by the Tshwane University of Technology apply.
2. **Ensure a single final version of your source code is handed in as requested.**
3. If needed, state all necessary assumptions clearly in code commentary.

MARKS: 100%

PAGES: 11 (incl. cover)

EXAMINER:

Mr A.J. Smith

Mr M. Chauke

Prof J.A. Jordaan

MODERATOR:

Mr D. Engelbrecht

TIME:

90 Minutes

15 Minutes extra time in the event of computer problems

FACULTY OF ENGINEERING AND THE BUILT ENVIRONMENT

DEPARTMENT OF ELECTRICAL ENGINEERING

ES216BB
ENGINEERING SOFTWARE DESIGN B

EVALUATION 3

OCTOBER 2026

EVALUATION INSTRUCTIONS

1. **Plagiarism:** Submit only original work. We will use similarity software to verify the authenticity of all submissions.

2. **Permitted Tools:** You are allowed to use only **CodeBlocks** and **Google Chrome** to access the evaluation, view the evaluation PDF and upload your submission for this evaluation. Access to emails, other online resources, and memory sticks is strictly prohibited. Please be aware that computer activity will be remotely monitored. Breaches of TUT's official examination and module rules will result in a minimum penalty of zero for this evaluation, with the potential for further disciplinary action.

3. **File Submission:** Your header code file must be named according to this format: **"<student number>.h" (e.g. 21011022.h).** Do not add any other text (name, surname, etc.) to the file name (ONLY YOUR STUDENT NUMBER).

4. **Uploading Instructions: ONLY SUBMIT YOUR HEADER FILE (.h)** via the designated upload link. While multiple uploads are allowed, only the most recent submission will be retained on the system. If you make an error in your initial upload, simply re-upload your file, and the previous version will be overridden.

5. **Evaluation Scope:** This assessment encompasses basic content from ES216AB and specifically ES216BB content defined in **Units 1 to 5**

6. **Programming Language:** Construct your program in **C++** and adhere to structured programming principles.

7. **Editing and Requirements:** Your program must meet all specified requirements. Refer to the attached appendices for additional details.

8. **Evaluation Requirements:**

    a. Remember to save your work on the PC "D: Drive" and save regularly throughout the evaluation.
    b. Do not modify the given code in the ".cpp" file except for changing the header file name to your student number, e.g. "123456789.h".
    c. Do not modify the given libraries and comments in the template ".h" file.
    d. Complete the C++ class definition and functions in each comment block as shown in the template ".h" file.Use the exact function names and parameters as used in the evaluation question paper and given ".cpp" file.

C++ FILE CODE EXPLANATION

The provided C++ file sets up a program that manages a laboratory equipment loan system. It uses a menu interface interacting with the **EquipmentLoan** class, which allows users to view and manage borrower loans for a specific piece of laboratory equipment.

The system includes functionalities for:

- **Displaying Laboratory Equipment Details**: Shows the equipment name, laboratory, capacity, and the number of current loans.
- **Loading Borrower Loans**: Reads borrower names from an external text file and adds them to the loan list.
- **Displaying Borrower List**: Prints all names of registered borrowers.
- **Displaying Loan Revenue**: Calculates total revenue based on loan fee and number of borrowers.

This system uses the **EquipmentLoan** class to encapsulate equipment-loan information, manage a dynamically allocated borrower list, and compute revenue from loans.

The required class definition and functions hereafter for the C++ header file should be implemented in the appropriate comment blocks as given in the .h template file.

CLASS DEFINITION EXPLANATION

The **EquipmentLoan** class defines the properties and methods required to manage the loan process:

- **Private Members:**
  - ***EquipmentName*** (**string**): Name of the laboratory equipment.
  - ***LaboratoryName (string):*** Name of the laboratory.
  - ***MaxBorrowers*** (**int**): Maximum number of borrowers allowed.
  - ***BorrowerList*** (**string***): Dynamically allocated array storing borrower names.
  - ***BorrowerCount*** (**int**): Tracks the number of current loans.

- **Public Functions:**
  - ***EquipmentLoan*****()**: Constructor.
  - ***~EquipmentLoan*****()**: Destructor.
  - ***SetDetails*****()**: Sets the equipment name, laboratory, and maximum borrowers.
  - ***AddBorrower*****()**: Registers a new borrower.
  - ***DisplayEquipmentDetails*****()**: Shows basic equipment-loan information.
  - ***DisplayBorrowerList*****()**: Displays all registered borrowers.
  - ***LoanRevenue*****()**: Calculates income from loans

CLASS FUNCTION EXPLANATIONS

1. Constructor Function

EquipmentLoan::EquipmentLoan()

- **Purpose:** Initialises member variables with safe defaults.
  - Strings are set to empty.
  - Integers to 0.
  - Pointers to nullptr.
- **Parameters:**
  - None.
- **Returns:**
  - Nothing (constructor).

---

2. Destructor Function

EquipmentLoan::~EquipmentLoan()

- **Purpose:** Deallocates the memory assigned to BorrowerList.
- **Parameters:**
  - None.
- **Returns:**
  - Nothing (destructor).

---

3. SetDetails Function

void EquipmentLoan::SetDetails(string EN, string LN, int MB)

- **Purpose:** Initializes the equipment-loan information.
- **Parameters:**
  - string EN – Equipment name.
  - string LN – Laboratory name.
  - int MB – Maximum number of borrowers.
- **Returns:**
  - Nothing.

---

4. AddBorrower Function

void EquipmentLoan::AddBorrower(string BorrowerName)

- **Purpose:** Adds a new borrower if capacity is available.
- **Parameters:**
  - string BorrowerName – Name of the borrower to register.
- **Returns:**
  - Nothing.

---

5. DisplayEquipmentDetails Function

void EquipmentLoan::DisplayEquipmentDetails()

- **Purpose:** Outputs the equipment name, laboratory, maximum capacity, and number of current loans.
- **Parameters:**
  - None.
- **Returns:**
  - Nothing. Outputs to console.

6. DisplayBorrowerList Function

void EquipmentLoan::DisplayBorrowerList()

- **Purpose:** Displays the list of registered borrowers or a message if no borrowers are registered.
- **Parameters:**
  - None.
- **Returns:**
  - Nothing. Outputs to console.

7. LoanRevenue Function

float EquipmentLoan::LoanRevenue(float LoanFee)

- **Purpose:** Computes total income from borrower loans.
  - Formula: BorrowerCount * LoanFee
- **Parameters:**
  - float LoanFee – The fee charged per borrower.
- **Returns:**
  - float value representing total revenue.

PRINT SCREENS

Main Menu:

```
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   Laboratory Equipment Loan System
   ES216BB
   Evaluation 3
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+

1. Display Equipment Details
2. Load Equipment Borrowers
3. Display Borrower List
4. Display Loan Revenue
5. Exit
Choice: 0
```

Display Equipment Details:

```
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   Laboratory Equipment Loan System
   ES216BB
   Evaluation 3
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+

1. Display Equipment Details
2. Load Equipment Borrowers
3. Display Borrower List
4. Display Loan Revenue
5. Exit
Choice: 1

Equipment Name: 3D Printer
Laboratory: Innovation Lab
Max Borrowers: 12
Current Borrowers: 0

Press any key to continue...
```

Load Equipment Borrowers:

```
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   Laboratory Equipment Loan System
   ES216BB
   Evaluation 3
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+

1. Display Equipment Details
2. Load Equipment Borrowers
3. Display Borrower List
4. Display Loan Revenue
5. Exit
Choice: 2

All possible borrowers loaded
Maximum borrowers reached: Mia not added.
Maximum borrowers reached: Ned not added.
```

Re-Display Equipment Details:

```
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   Laboratory Equipment Loan System
   ES216BB
   Evaluation 3
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+

1. Display Equipment Details
2. Load Equipment Borrowers
3. Display Borrower List
4. Display Loan Revenue
5. Exit
Choice: 1

Equipment Name: 3D Printer
Laboratory: Innovation Lab
Max Borrowers: 12
Current Borrowers: 12

Press any key to continue...
```

Display Borrower List:

```
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   Laboratory Equipment Loan System
   ES216BB
   Evaluation 3
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+

1. Display Equipment Details
2. Load Equipment Borrowers
3. Display Borrower List
4. Display Loan Revenue
5. Exit
Choice: 3

Borrower List: Ava ; Ben ; Cia ; Dan ; Eli ; Fay ;
               Gus ; Hope ; Ira ; Jade ; Kai ; Lia ;

Press any key to continue
```

Display Loan Revenue:

```
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   Laboratory Equipment Loan System
   ES216BB
   Evaluation 3
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+

1. Display Equipment Details
2. Load Equipment Borrowers
3. Display Borrower List
4. Display Loan Revenue
5. Exit
Choice: 4

Loan Revenue: R3000

Press any key to continue...
```

Exit Application:

```
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   Laboratory Equipment Loan System
   ES216BB
   Evaluation 3
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+

1. Display Equipment Details
2. Load Equipment Borrowers
3. Display Borrower List
4. Display Loan Revenue
5. Exit
Choice: 5

Exit Program...

Process returned 0 (0x0)

Press any key to continue...
```

ANNEXURE A – MARK ALLOCATION

Note: Score range is 0 - 4 which is: 0-none, 1-poor, 2-average, 3-good, 4-excellent

TEST RUBRIC	SCORE [0-4]	WEIGHT [%]
C++ CODE EVALUATION		**50+2**
0. Class Definition		**6**
1. Class Constructor Function		**4**
2. Class Destructor Function		**4**
3. Set Details Function		**5**
4. Add Function		**8**
5. Display Details Function		**5**
6. Display List Function		**5**
7. Profit Function		**5**
8. Overall Impression		**5**
9. No Compile or Runtime Errors		**5**
TOTAL		**50**

Graduate Attribute	GA Number	GA Score [0-5]
Application of scientific and engineering knowledge	GA2	4,7
Engineering methods, skills, tools, including information technology	GA5	0,1,2,3
Impact of Engineering Activity	GA7	5,6
Engineering Professionalism	GA10	8,9

ANNEXURE B – INFORMATION SHEET

Data types: void, char, short, int, float, double
Data Type modifiers: const, auto, static, unsigned, signed
Arithmetic operators: * / % + -
Relational operators: < <= > >= == !=
Assignment operator: = += -= *= /= %= &= ^= |= <<= >>=
Logic operators: && || !
Bitwise logic operators: & | ^ ~ << >>
Pointer operators: Derefernce: * Address: &
Control Structures:

IF Selection: if (condition) { … };
IF ELSE Selection: if (condition) { … } else { … };
WHILE Loop: while (condition) { … };
DO WHILE loop: do { … } while (condition);
FOR Loop: for (initial value of control variable; loop condition; increment of control variable) { … }
SWITCH Selection: switch (control variable){ case 'value': … ; break; default: … ; break; }

Functions: return_data_type function_name (parameters) { … };
Common Library Functions: printf() , scanf() , rand() , srand() , time() , isalpha() , isdigit() , getchar() , getch(), strcpy()
Arrays:

One dimensional: data_type variable_name[size];
Two dimensional: data_type variable_name [x_size][y_size];

ANNEXURE C – ASCII TABLE

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040		Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174			|
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177	&#127;	DEL

Source: www.LookupTables.com